

## Solutions: Laplace Transforms

Table references are to the Laplace pairs and properties handout (equivalently, Nise Tables 2.1 and 2.2). Every  $f(t)$  below is understood to be multiplied by the unit step  $u(t)$ . Numerical verification of Parts 2–4 is in `assignment_solutions.m`.

### Part 1: Forward Laplace Transform

1.  $f(t) = 4e^{-3t}$ . Exponential pair with  $a = 3$ , scaled by 4:

$$F(s) = \frac{4}{s+3}$$

2.  $f(t) = 2\sin(5t)$ . Sine pair with  $\omega = 5$ :

$$F(s) = \frac{2 \cdot 5}{s^2 + 25} = \frac{10}{s^2 + 25}$$

3.  $f(t) = 3\cos(2t)$ . Cosine pair with  $\omega = 2$ :

$$F(s) = \frac{3s}{s^2 + 4}$$

4.  $f(t) = te^{-2t}$ . Ramp pair  $t \leftrightarrow 1/s^2$ , then the frequency-shift property  $e^{-at}f(t) \leftrightarrow F(s+a)$  with  $a = 2$ :

$$F(s) = \frac{1}{(s+2)^2}$$

5.  $f(t) = 5 - 5e^{-t}$ . Linearity: transform each term separately.

$$F(s) = \frac{5}{s} - \frac{5}{s+1} = \frac{5(s+1) - 5s}{s(s+1)} \Rightarrow F(s) = \frac{5}{s(s+1)}$$

Either form is acceptable. The combined form shows the two poles at  $s = 0$  and  $s = -1$  that generate the constant and the decaying exponential.

### Part 2: Inverse Laplace Transform by Partial Fractions

6.  $F(s) = \frac{3}{s(s+3)}$ . Two distinct real poles:

$$\frac{3}{s(s+3)} = \frac{A}{s} + \frac{B}{s+3}, \quad A = \left. \frac{3}{s+3} \right|_{s=0} = 1, \quad B = \left. \frac{3}{s} \right|_{s=-3} = -1.$$

$$f(t) = 1 - e^{-3t}$$

$$7. F(s) = \frac{s+1}{(s+2)(s+4)}.$$

$$\frac{s+1}{(s+2)(s+4)} = \frac{A}{s+2} + \frac{B}{s+4}, \quad A = \left. \frac{s+1}{s+4} \right|_{s=-2} = \frac{-1}{2}, \quad B = \left. \frac{s+1}{s+2} \right|_{s=-4} = \frac{-3}{-2} = \frac{3}{2}.$$

$$\boxed{f(t) = -\frac{1}{2}e^{-2t} + \frac{3}{2}e^{-4t}}$$

Check:  $f(0^+) = -\frac{1}{2} + \frac{3}{2} = 1$ , which matches the initial-value theorem,  $\lim_{s \rightarrow \infty} sF(s) = 1$ .

$$8. F(s) = \frac{2}{s^2 + 4s + 3}. \text{ Factor the denominator first: } s^2 + 4s + 3 = (s+1)(s+3).$$

$$\frac{2}{(s+1)(s+3)} = \frac{A}{s+1} + \frac{B}{s+3}, \quad A = \left. \frac{2}{s+3} \right|_{s=-1} = 1, \quad B = \left. \frac{2}{s+1} \right|_{s=-3} = -1.$$

$$\boxed{f(t) = e^{-t} - e^{-3t}}$$

$$9. F(s) = \frac{10}{s(s^2 + 4s + 8)}. \text{ The quadratic has complex roots } (s = -2 \pm 2j), \text{ so keep it as a quadratic factor:}$$

$$\frac{10}{s(s^2 + 4s + 8)} = \frac{A}{s} + \frac{Bs + C}{s^2 + 4s + 8}.$$

Cover-up gives  $A = 10/8 = 5/4$ . Clearing denominators,

$$10 = A(s^2 + 4s + 8) + (Bs + C)s,$$

and matching coefficients:  $s^2$ :  $A + B = 0 \Rightarrow B = -5/4$ ;  $s^1$ :  $4A + C = 0 \Rightarrow C = -5$ . So

$$F(s) = \frac{5/4}{s} - \frac{5}{4} \cdot \frac{s+4}{s^2 + 4s + 8}.$$

Complete the square,  $s^2 + 4s + 8 = (s+2)^2 + 2^2$ , and split the numerator so that it matches the shifted cosine and sine pairs:

$$\frac{s+4}{(s+2)^2 + 2^2} = \frac{s+2}{(s+2)^2 + 2^2} + \frac{2}{(s+2)^2 + 2^2} \leftrightarrow e^{-2t} \cos(2t) + e^{-2t} \sin(2t).$$

$$\boxed{f(t) = \frac{5}{4} \left[ 1 - e^{-2t} (\cos 2t + \sin 2t) \right]}$$

Checks:  $f(0^+) = \frac{5}{4}(1-1) = 0$  and  $f(\infty) = \frac{5}{4} = \lim_{s \rightarrow 0} sF(s) = 10/8$ . Using  $\cos \theta + \sin \theta = \sqrt{2} \cos(\theta - \pi/4)$ , the same result can be written  $f(t) = \frac{5}{4} [1 - \sqrt{2} e^{-2t} \cos(2t - \pi/4)]$ .

### Part 3: Solving ODEs via Laplace Transforms

Zero initial conditions, so  $\mathcal{L}\{\dot{c}\} = sC(s)$  and  $\mathcal{L}\{\ddot{c}\} = s^2C(s)$ . The input is  $R(s) = 1/s$ .

10.  $\dot{c} + 4c = 8r$ .

(a)  $sC(s) + 4C(s) = 8R(s)$ .

(b)  $C(s) = \frac{8}{s+4} \cdot \frac{1}{s} = \frac{8}{s(s+4)}$ .

(c)  $\frac{8}{s(s+4)} = \frac{2}{s} - \frac{2}{s+4}$ , so

$$c(t) = 2(1 - e^{-4t})$$

The response rises to a final value of 2 (the DC gain  $8/4$ ) with time constant  $\tau = 1/4$  s.

11.  $\ddot{c} + 5\dot{c} + 6c = 6r$ .

(a)  $(s^2 + 5s + 6)C(s) = 6R(s)$ .

(b)  $C(s) = \frac{6}{s(s^2 + 5s + 6)} = \frac{6}{s(s+2)(s+3)}$ .

(c) Three distinct real poles:

$$\frac{6}{s(s+2)(s+3)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+3},$$

$$A = \left. \frac{6}{(s+2)(s+3)} \right|_{s=0} = 1, \quad B = \left. \frac{6}{s(s+3)} \right|_{s=-2} = \frac{6}{(-2)(1)} = -3, \quad C = \left. \frac{6}{s(s+2)} \right|_{s=-3} = \frac{6}{(-3)(-1)} = 2.$$

$$c(t) = 1 - 3e^{-2t} + 2e^{-3t}$$

Checks:  $c(0) = 1 - 3 + 2 = 0$  and  $\dot{c}(0) = 6 - 6 = 0$ , consistent with the zero initial conditions. The response is overdamped (two real poles) and settles at the DC gain  $6/6 = 1$ .

12.  $\ddot{c} + 4\dot{c} + 8c = 8r$ .

(a)  $(s^2 + 4s + 8)C(s) = 8R(s)$ .

(b)  $C(s) = \frac{8}{s(s^2 + 4s + 8)}$ .

(c) This is  $8/10$  times the expression in Problem 9, so by linearity

$$c(t) = \frac{8}{10} \cdot \frac{5}{4} \left[ 1 - e^{-2t}(\cos 2t + \sin 2t) \right] = 1 - e^{-2t}(\cos 2t + \sin 2t).$$

Writing  $\cos 2t + \sin 2t = \sqrt{2} \cos(2t - \pi/4)$  puts it in the requested form  $c(t) = K[1 - A e^{-\sigma t} \cos(\omega_d t - \phi)]$ :

$$c(t) = 1 - \sqrt{2} e^{-2t} \cos\left(2t - \frac{\pi}{4}\right) \quad K = 1, A = \sqrt{2}, \sigma = 2, \omega_d = 2, \phi = \pi/4.$$

The poles are  $s = -2 \pm 2j$ , so  $\sigma$  is the real part and  $\omega_d$  the imaginary part. In standard second-order form,  $\omega_n = \sqrt{8}$  and  $\zeta = 4/(2\sqrt{8}) = 1/\sqrt{2} \approx 0.707$ : underdamped, with the oscillation decaying inside the envelope  $\sqrt{2} e^{-2t}$ .

**Part 4: Pole Locations and Time-Domain Behavior**

13. Poles and step-response character for each system:

|     | $G(s)$                   | Poles       | Stability / step response                                                                                                                   |
|-----|--------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| (a) | $\frac{1}{(s+1)(s+3)}$   | $-1, -3$    | Stable. Overdamped: sum of two decaying exponentials, no overshoot.                                                                         |
| (b) | $\frac{1}{s^2 + 2s + 5}$ | $-1 \pm 2j$ | Stable. Underdamped ( $\zeta = 1/\sqrt{5} \approx 0.45$ ): decaying oscillation with overshoot.                                             |
| (c) | $\frac{1}{s(s+2)}$       | $0, -2$     | Marginally stable (pole at the origin). Step response grows without bound: after the $e^{-2t}$ transient dies out it ramps at slope $1/2$ . |
| (d) | $\frac{1}{(s-1)(s+3)}$   | $+1, -3$    | Unstable (right-half-plane pole). Step response grows exponentially as $e^t$ .                                                              |

For (c), the pole at the origin is an integrator:  $C(s) = 1/(s^2(s+2))$  has a double pole at  $s = 0$ , giving  $c(t) = \frac{1}{2}t - \frac{1}{4} + \frac{1}{4}e^{-2t}$ . The system is not asymptotically stable, but the divergence is a ramp rather than an exponential, and there is no oscillation.

14. MATLAB. Each system is a `tf` object; `step()` on a stable system runs until the response settles, so for (c) and (d) pass an explicit time vector to see the growth.

```
s = tf('s');
Ga = 1/((s+1)*(s+3));
Gb = 1/(s^2 + 2*s + 5);
Gc = 1/(s*(s+2));
Gd = 1/((s-1)*(s+3));

t = 0:0.01:6;
figure(1); clf
subplot(2,2,1); step(Ga, t); title('(a) overdamped')
subplot(2,2,2); step(Gb, t); title('(b) underdamped')
subplot(2,2,3); step(Gc, t); title('(c) pole at origin: ramps')
subplot(2,2,4); step(Gd, t); title('(d) unstable: e^t growth')

pole(Ga), pole(Gb), pole(Gc), pole(Gd)
```

Expected plots: (a) a smooth monotonic rise to  $1/3$ ; (b) an oscillatory rise to  $1/5$  with about 20 % overshoot; (c) a curve that becomes a straight line of slope  $1/2$ ; (d) a curve that bends upward without limit. The `pole()` output confirms the locations in the table.